

NOVA

IMS

Information
Management
School

MDSAA

Master Degree Program in
Data Science and Advanced Analytics

Business Cases with Data Science

Case 2: Hierarchical Forecasting – Siemens Advanta Consulting

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April, 2026

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1. EXECUTIVE SUMMARY

This project was developed for the Siemens Advanta Consulting business case and addresses the need for a highly accurate and objective financial forecasting system. The primary goal was to transition from subjective estimation methods toward an automated machine learning framework capable of predicting revenue (the target) across all hierarchical levels given in data. The methodology uses a bottom-up forecasting approach. Data preparation involved rigorous consistency checks, such as the removal of inconsistent subsegments and advanced feature engineering. With time series techniques, key predictors were created, such as lags, rolling averages to capture seasonality and differencing to identify economic acceleration or volatility. Furthermore, an imputation strategy was applied, distinguishing between stationary and non-stationary series to ensure that missing values were all filled. The modeling phase contained different algorithms, including LightGBM, XGBoost, Random Forest and Long Short-Term Memory (*LSTM*). Evaluation focused on a Weighted Root Mean Squared Error (*RMSE*) metric, which prioritizes accuracy in high revenue segments to ensure the most significant financial impacts are predicted with the highest precision. The final weighted ensemble model effectively captures market trends and seasonalities, providing a reliable baseline for global strategic planning.

2. BUSINESS NEEDS AND REQUIRED OUTCOME

2.1. Business Objectives

The main goal of this project is to build a reliable model that predicts our revenue with high accuracy. By leveraging historical data and machine learning, we were able to move away from manual estimators toward a more objective, automated planning process. This approach is designed to improve financial predictability by providing accurate revenue expectations, which allows for better informed decisions. Furthermore, the project focuses on ensuring organizational alignment by creating a single source of truth where forecasts at the most detailed levels align with high level global targets. This consistency ensures all are working toward the same goals while increasing operational efficiency. By identifying the most important drivers of revenue, we reduce data complexity and make insights easier for stakeholders to act upon.

2.2. Business Success Criteria

The success of this project is measured by its ability to provide a stable and trustworthy forecast that the business can use for planning. First, the model must achieve high accuracy, ensuring that the gap between predicted and actual revenue is kept to minimum. Second, the system must maintain total hierarchical consistency. Another key for success is the simplification of our data process. By identifying the most important features, we create a model that is easier to explain to stakeholders and faster to update as new data arrives but also with less probability of leading to overfitting. Finally, success is defined by providing leadership with a reliable single source that reduces uncertainty and allows for more confident budgeting and resource allocation across all business units.

2.3. Situation assessment

The situation assessment identified that managing such a large volume of data across multiple business levels was a significant challenge. With many different variables involved, it became difficult to ensure consistency at every phase of the process. This complexity made it harder to identify which specific factors were truly driving revenue and which were simply noise that did not serve the project's objectives well. During the project, we evaluated the available historical data to find a more efficient way to connect these different levels. By observing the data structure, we determined that focusing on a smaller, more impactful set of features would create a more stable and accurate model for the business.

2.4. Determine Data Mining goals

The main objective of the project is to predict the future revenue volume, designed as *Revenue cons. (anon)*, generated by the different business sub-segments of Siemens Advanta over time. As demonstrated in the model preparation phase, the variable related to orders (*Orders cons. (anon)*) was explicitly removed to focus the analysis exclusively on *Revenue*.

This is a Time Series Forecasting problem that needs to be solved using regression algorithms (such as LightGBM and XGBoost). As this is a complex problem, the chosen modeling approach was designed to handle the diverse nature of the business's hierarchical structure.

The success of this objective is measured by the model's ability to ensure accurate hierarchical consolidation, capture market trends (growth/decline) and seasonalities, generalizing the knowledge to new periods with the highest possible accuracy. The metric chosen to evaluate the business objective and model effectiveness is the *RMSE*, calculated on an absolute basis for each business level assigning greater weight to higher revenues. This error is then normalized and consolidated using a structured impact distribution of 25% for Sub-segment level, 25% for Segment level and 50% for Business Unit.

The ultimate data mining objective is, therefore, to configure the optimal model to minimize this specific evaluation metric on the validation set.

3. METHODOLOGY

3.1. Data understanding

In this initial phase, the datasets were explored to understand their structure, information quality and the relationships between variables. This step is crucial for identifying inconsistencies and planning preprocessing needs.

3.1.1. Market Dataset

The *market* dataset provides external macroeconomic context by capturing monthly macroeconomic indicators from key global economies.

The data is structured to analyze how global market trends, regional economic health and industrial outputs might influence the commercial performance of the organization.

Market Dataset Metadata

The dataset contains the following informational fields:

Period: A discrete temporal identifier that maps to specific calendar months and years. This serves as the primary key to merge the macroeconomic data with the internal sales periods.

Date: The exact calendar month and year corresponding to each Period identifier.

[Country]_Exports: Represents the volume of goods exported by each respective country, indicating the health of their international trade and manufacturing demand.

[Country]_GDP: Represents the Gross Domestic Product for the 8 selected global markets (China, France, Germany, Italy, Japan, Switzerland, United Kingdom and United States).

This includes total economic output and specific sector breakdowns, explicitly named as *[Country]_GDP_from_Construction* and *[Country]_GDP_from_Manufacturing*.

[Country]_Industrial_Production and *[Country]_Industrial_Production_Mom*: Tracks manufacturing health via broader industrial production indices and their month-over-month evolution.

[Country]_Inflation_Rate and *[Country]_Core_Inflation_Rate*: Captures pricing trends and the cost of living in each respective economy through general and core inflation rates.

[Country]_Interest_Rate: Represents the benchmark borrowing costs set by national central banks, providing vital context on regional investment behaviors.

[Country]_Steel_Production: Represents a highly specific industrial output metric for each analyzed region.

3.1.2. Train Dataset

The *train* dataset captures daily sales data from Siemens product groups.

The data is structured to analyze commercial performance across different organizational levels, i.e., Business Units, Business Segments and Business Subsegments with a focus on orders and revenue.

Train Dataset Metadata

The dataset contains the following informational fields:

Anon Period: A temporal identifier indicating the specific time period (e.g., month and year) of each transaction. This allows for time-series analysis and period-over-period comparisons.

TGL Biz Desc: A high-level business description code for the *Siemens Smart Infrastructure* (SSI) historical sales.

TGL Business Unit: An identifier for a specific operational unit within the organization. This represents a mid-level aggregation of business activities.

TGL Business Segment: A more specific breakdown within the Business Unit, indicating a

specific product or service category.

TGL Business Subsegment: The most detailed hierarchical level, providing a finer classification of the business activity.

Orders cons. (anon): Represents customer orders.

Revenue cons. (anon): Represents the revenue generated.

3.2. Data preparation

3.2.1. Train Data

Consistency Checking

To ensure the reliability of model evaluation, we performed a categorical integrity check between the training and validation sets. This step verifies that all categorical values present in the validation set were also observed during training, preventing evaluation failures caused by unseen categories.

For each categorical column (specifically *TGL Business Unit*, *TGL Business Segment*, and *TGL Business Subsegment*), we compared the unique values between the training and validation sets.

The *TGL Business Unit* and *TGL Business Segment* columns showed complete alignment between training and validation sets, with no new categories introduced in the validation data.

However, the *TGL Business Subsegment* column contained one value present in the validation set that was absent from the training set, which was the subsegment *SSI0278218*.

The inconsistent subsegment identifier was removed from the validation set to maintain data integrity and ensure valid model evaluation.

To facilitate consistent data handling across all processing stages, we standardized the column naming convention.

The column originally named *Anon Period* was renamed to *Period* in both the training and validation datasets.

Our target variable was *Revenue*, with *Orders* available as a potential feature or auxiliary output.

Taking that into account, we removed the columns *Revenue* and *Orders* from the test set to prevent data leakage, ensuring that the model does not have access to the real values during prediction.

Data Type Verification

The *China_Steel_Production* column was cast to *float64* to preserve precision

Missing Values Handling

For supervised learning, missing values were identified and addressed prior to model training.

The *train* dataset does not contain any missing values.

The missing entries occur identically across both columns, suggesting systematic absence (e.g., no transaction recorded, data collection gaps, or filtered records).

Duplicates Detections

There were no duplicate rows in any of the datasets and no multiple entries sharing the same Business Unit, Subsegment and Period.

3.2.2. Market Data

Missing Values Handling

The market dataset showed varying degrees of missingness across multiple columns, with steel production and interest rate columns showing the highest incompleteness. GDP figures for major economies exhibit a consistent missing pattern (8 observations each).

A two-stage imputation pipeline was implemented to address missing values in the market dataset.

The dataset was first sorted chronologically by *Period* to establish temporal order.

For all columns except *Period*, missing values were filled using forward fill, carrying the last observed value forward. Forward fill respects temporal dependencies and is appropriate for time-series data.

After forward fill, linear interpolation was applied to any remaining missing values.

The interpolation estimates values based on the trend between neighboring observed points. Linear interpolation provides a more complex estimate than simple forward fill for isolated gaps, capturing underlying trends in the data.

Duplicates Detections

There were no duplicate rows in the dataset.

Outliers

Outlier analysis is essential for machine learning preprocessing to identify potentially influential data points that could distort model training.

Having this in mind, we employed the Interquartile Range (*IQR*) method, defining outliers as values falling outside the range.

The variables with the most outliers were *Japan_GDP* (64 outliers) and *United_Kingdom_Interest_Rate* (55 outliers) being the most extreme.

In conclusion, there was not a significant number of outliers in the dataset.

3.3. Feature Engineering

Based on the variables from each country regarding the different industrial and steel productions, as well as interest rate, the variables *Global_Industrial_Production*, *Global_Steel_Industrial_Production* and *Global_Interest_Rate* were created.

All features were standardized to zero mean and unit variance, to ensure that the features are on a similar scale, which can help improve the performance of machine learning algorithms that are sensitive to the scale of the data.

Also, to capture the economic condition of each country in a single, informative feature, we created new variables that were calculated by weighting each variable for every country.

The resulting variables provide a compact, interpretable representation of economic conditions across countries.

Moreover, to improve model stability and interpretability, we identified and removed highly correlated features from the dataset, since high multicollinearity can negatively impact certain machine learning models and lead to redundant information being weighted multiple times.

The absolute correlation matrix was computed for all numerical variables in the dataset and extracted all variable pairs with a correlation coefficient exceeding 0.95.

Finally, to capture temporal dynamics, constructed lagged values (periods 1, 3, 6, and 12), period-over-period differences (1 and 3 periods), rolling means (3 and 6 periods), and rolling standard deviation (3 periods) for each key variable were added. All rolling statistics include a one-period shift to prevent data leakage, ensuring that only past information is used for prediction at any given time point.

Lagged features were created at four distinct intervals: 1, 3, 6, and 12 periods. A lag of 1 period provides the model with the immediately preceding observation.

The lag of 3 periods captures the value from approximately one quarter ago, while the 6-period lag represents a half-year historical perspective. The 12-period lag enables annual comparisons.

Difference features help the model understand whether the economy is accelerating or decelerating, independent of whether current values are high or low.

Rolling mean features allowed smoothing out short-term fluctuations and reveal underlying trends.

Rolling standard deviation captures the variability or instability of a variable over time.

3.4. Merge Train and market Dataset Processing

For the training phase, the original train and test dataset were merged with the new external market data, using the *Period* feature as the key. This integration ensured that each observation was enriched with relevant market information, allowing the model to capture both internal and external factors influencing the target variable.

After the merge, the final dataset consisted of 4237 observations for training and 712 observations for testing.

To ensure data quality, consistency and improvement of our project, several preprocessing steps were applied.

Outliers

In this project, no explicit outlier removal was performed. This decision was made deliberately based on the nature of the data and the modeling approach. In time series data, particularly in this case, *Revenue* and macroeconomic contexts, often exhibit extreme values that correspond to real-world events, such as demanding spikes, economic shocks or seasonal effects. Removing them would lead to the loss of important information and reduce model ability to capture meaningful patterns.

Feature Engineering

Since it was a time series problem, and bottom-up approach was chosen, the first step for the feature engineering was the analysis of stationary for the *TGL Business Subsegment*, which had resulted and the new binary feature *stationary* where 1 represented the stationary subsegments. For the segments that had not enough observations we assumed that they were not stationary and for the consistent series we assumed they were.

The first group of engineered features consisted of lag variables. Specifically, lag values from 1 to 4 periods were created, resulting in the features *Revenue_lag_1*, *Revenue_lag_2*, *Revenue_lag_3*, and *Revenue_lag_4*. These variables represent past revenue values and allow the model to capture temporal dependencies, as current revenue is often influenced by recent historical observations. Lag features are particularly relevant in time series forecasting because they help the model learn recurring patterns, persistence effects, and short-term dynamics.

To further account for changes in the series, difference features were also introduced. The features *Revenue_diff_1* and *Revenue_diff_3* measure the variation in revenue compared to the previous period and to the value three periods earlier, respectively. These transformations help the model focus on the direction and magnitude of change rather than only on absolute revenue levels. In addition, differencing is especially useful for handling non-stationary series, as it reduces the impact of trends and makes the data more stable over time.

Finally, rolling statistics were created to provide local context on recent revenue behavior. The feature *Revenue_roll_mean_3* captures the average revenue over the previous three periods, while *Revenue_roll_std_3* measures the standard deviation over the same *Period*. Both features were computed using shifted values to ensure that only past information was used. The rolling mean helps smooth short-term fluctuations and highlights local trends, whereas the rolling standard deviation captures the degree of variability or volatility in recent periods.

By including past values of *Revenue*, the model was able to learn from previous behavior, improving its ability to identify trends and generate more accurate forecasts.

Missing values handling

For the missing values created by the Lags features two approaches were followed. For subsegments classified as stationary, missing values were filled using linear interpolation, complemented by forward fill and backward fill. This approach is appropriate for stationary series since it preserves their relatively stable behavior over time. For non-stationary series, a different strategy was adopted. Missing values in lag features were filled by estimating a linear trend based on the observed values of the series, allowing the imputed values to follow the overall direction of change. When there were not enough observations to estimate a trend reliably, forward fill and backward fill were used instead.

For rolling features, stationary series were again imputed using interpolation plus forward/backward filling, while non-stationary series were filled only with forward fill and backward fill to avoid introducing artificial patterns.

Overall, this approach ensured that missing values were treated in a way that was consistent with the temporal behavior of each series, preserving both stability in stationary subsegments and trend information in non-stationary ones.

3.5. Feature Selection for Modeling

In terms of feature selection for modeling, an initial step was performed to improve model efficiency and reduce redundancy, where highly correlated features were identified and removed prior to model training.

This approach helps mitigate multicollinearity issues, reduces noise in the dataset, and ensures that the model is not biased by duplicated information, ultimately contributing to more stable and interpretable results.

4. MODELING

4.1. Train-Test Split

Dataset was split into training (periods 1 to 36) and validation sets (37 to 38), showed on the graph below (*Figure 1*), to evaluate the model's ability to predict unseen data.

These variations indicate that past behavior influences future values, which is a key characteristic of time series data. By splitting the data in March 2024 (period 36), ensuring that the validation set reflects the most recent patterns, allowing us to capture changes over time and evaluate how well the model adapts to them.



Figure 1 - Train vs Validation Split for Modeling.

4.2. Data Preprocessing for Modeling

To ensure suitable format for modeling the data was cleaned and transformed. For each entry it was set a unique identifier by combining multiple columns (*Period*, *TGL Business Unit*, *TGL Business Segment*, *TGL Business Subsegment*).

4.3. Model Selection and Hyperparameter Tuning

4.3.1. LightGBM

The first model tested was *LightGBM*, an efficient gradient boosting model, to predict the target variable. A grid search was performed to find the best hyperparameters, such as the number of trees (*n_estimators*), learning rate, and number of leaves in the tree.

The results indicated that the model predicts with an error of approximately 3.5 million euros on average.

4.3.2. XGBoost Model

XGBoost, a powerful machine learning model, improve the accuracy of revenue predictions. This model allows us to handle more complex relationships in the data, improving forecast reliability. To optimize the model's performance, several key hyperparameters were tuned, ending up with the the best model configuration, delivering a highly optimized prediction.

The results indicated that the model predicts with an error of approximately 2.9 million euros on average, with the RMSE on the validation set significantly lower than the previous model.

Feature Selection for XGBoost

To improve model accuracy feature selection was performed, by removing irrelevant or noisy features. The model showed a stable performance, with a small gap between train and validation errors.

4.3.3. Random Forest Model

The Random Forest algorithm, a powerful model known for handling non-linear relationships and reducing overfitting by aggregating multiple decision trees was applied. While Random Forest delivered a higher *RMSE* compared to XGBoost and LightGBM, it remains a strong option for models requiring flexibility across complex, non-linear patterns. However, the model could benefit from further optimization since it indicates potential overfitting.

4.3.4. Weighted Ensemble

To further enhance model performance, weighted combination of predictions was made from LightGBM (5%), XGBoost (55%), selected features XGBoost (30%) and Random Forest (10%), into a Weighted Ensemble. This approach helps mitigate the weaknesses of individual models and boosts overall accuracy.

The Weighted Ensemble approach successfully reduced the error by approximately 1 million euros compared to the individual models, showcasing the value of combining different models for more reliable predictions.

The validation graph for the ensemble model shows a much more stable and accurate prediction trajectory compared to previous models. The ensemble model (*Figure 2*) now tracks real revenue closely across the entire validation period.

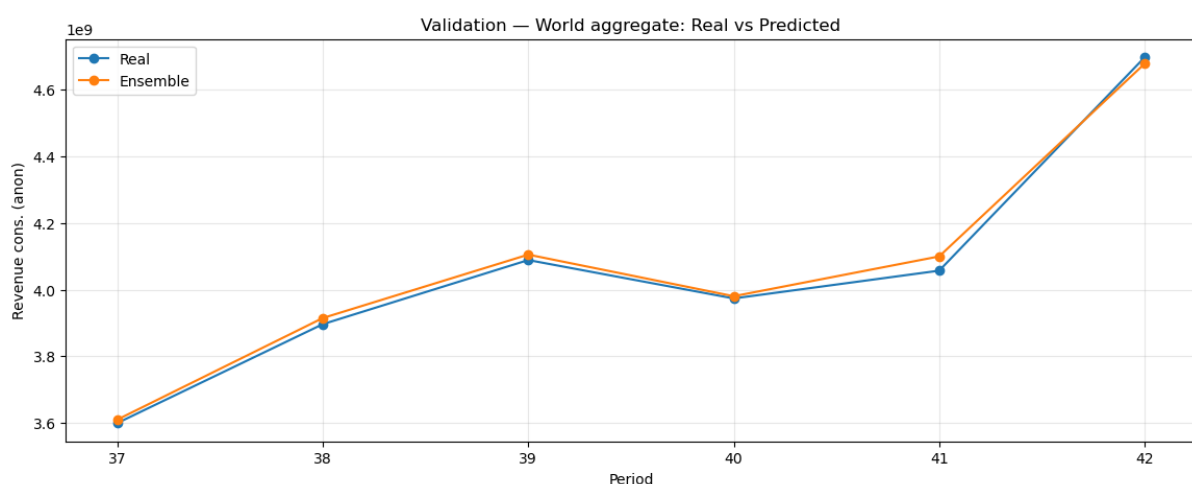


Figure 2 - Real vs Predicted for Ensemble model.

5. RESULTS EVALUATION

The table below summarizes the final error and *RMSE* for each model tested:

Table 1 - Model Performance Comparasion

Model	Final Error	RMSE
LightGBM	0.04197	3 537 926€
XGBoost	0.03459	2 970 130.22€
XGBoost v2	0.04019	3 444 641.16€
Random Forest	0.06164	5 089 071.76€
Ensemble (Best)	0.03321	2 796 580.11€

The clear winner is the Weighted Ensemble, which combines multiple models, delivering the lowest final error and the best *RMSE*.

In conclusion, the Weighted Ensemble model demonstrated the most reliable and accurate predictions across all business subsegments, with an *RMSE* of 2.8 million euros. This model combines multiple algorithms (LightGBM, XGBoost, and Random Forest) and capitalizes on their individual strengths, reducing the errors that single models experienced.

Moving forward, further model optimization and hyperparameter fine-tuning could improve results even more, especially in handling edge cases where individual models may struggle.

6. DEPLOYMENT AND MAINTENANCE PLANS

The deployment phase will focus on automating the end-to-end data pipeline, continuously integrating both internal historical revenue records and external macroeconomic indicators. The final weighted ensemble model will be integrated into the organization's central planning systems to automatically generate bottom-up forecasts. By making these automated predictions, the business will achieve its goal of establishing a single source of truth for all stakeholders, enabling objective and confident budgeting decisions across all hierarchical levels.

The final forecasting model will operate as an automated system that generates monthly revenue predictions using both internal sales data and external market indicators. The solution will be integrated into the company's existing planning tools, allowing stakeholders to easily access forecasts and use them for decision-making.

The deployment process includes a data pipeline that automatically collects and preprocesses new data, applies the same feature engineering steps used during training, and feeds it into the model to produce predictions. These forecasts are generated at the most detailed level (subsegment) and then aggregated to ensure consistency across segments and business units.

To support business needs, the model will serve as a single source of truth for financial planning, helping improve budgeting, resource allocation, and overall forecasting accuracy.

By replacing manual estimation processes, the system increases efficiency and reduces subjectivity.

To ensure reliability over time, the model's performance will be continuously monitored using the Weighted *RMSE* metric. Predicted values will be compared with actual results to detect any decrease in accuracy. Additionally, changes in data patterns will be tracked to identify potential data drift.

The model will be regularly updated to reflect new data and evolving market conditions. Retraining will be performed periodically, or more frequently if significant changes occur. Feature relevance will also be reviewed to maintain model simplicity and avoid overfitting.

Finally, proper documentation and version control will be maintained to ensure transparency and traceability. This guarantees that the system remains robust, reliable, and aligned with business objectives over time.

7. CONCLUSIONS

In this project, we focused on forecasting revenue for Siemens by analyzing various business units and segments, with a particular emphasis on improving accuracy and reliability. We used a combination of machine learning models, including LightGBM, XGBoost, and Random Forest, and evaluated their performance on time series data to predict future revenue trends.

Through a bottom-up hierarchical forecasting approach, we generated granular insights from the lowest business levels and aggregated them to higher levels. This methodology allowed us to capture a more detailed picture of the business's revenue dynamics, leading to better forecasts.

The Weighted Ensemble model emerged as the most accurate and reliable solution, outperforming individual models by significantly reducing error. This result underlines the power of combining different models to improve forecasting accuracy, ultimately supporting better business decision-making.

In particular, the integration of the model into Siemens existing systems will enable the company to make data-driven decisions in real-time, optimizing revenue prediction and business strategies. By utilizing predictive analytics, Siemens can enhance resource allocation, plan strategically, and boost profitability across its business units.

Overall, this project demonstrates how advanced data analytics can improve forecasting accuracy, business strategy alignment, and decision-making processes, ultimately driving both growth and operational efficiency within the company.

7.1. Considerations for model improvement

To improve the performance and robustness of the model, several strategies can be explored. First, increasing the size and diversity of the training dataset could help the model generalize better and reduce overfitting.

Second, incorporating methods to better capture seasonality and long-term trends. This could involve, for example, using techniques such as lag features, rolling statistics, or differencing from the previous month's revenue to improve the quality of the input data before modeling.

Finally, experimenting with different model architectures may yield better results. For instance, using ensemble methods to combine predictions from multiple models can improve performance by utilizing the advantages of many algorithms while mitigating their weaknesses.